

WealthSphere

AI-Powered Portfolio & Wealth Management Platform

Detailed Project Brief

Android + iOS Mobile Application | Portfolio Analytics | Compound Growth
Forecasting | AI Investment Intelligence

Purpose: Define the product, architecture, functional modules, AI design, security model and phased implementation approach for a cloud-ready personal wealth intelligence application.

Version 1.0 - September 2026

1. Executive Summary

WealthSphere is a mobile-first personal wealth intelligence platform for Android and iOS. It consolidates multiple investment vehicles into one portfolio view, measures performance, tracks passive income and net worth, models compound growth, manages financial goals, and uses an AI Investment Intelligence layer to help users research possible investment opportunities using current market evidence, portfolio context and transparent risk analysis.

The product is designed to move beyond a basic portfolio tracker. Its long-term role is to become a personal wealth operating system: a trusted record of assets and transactions, a deterministic financial analytics engine, a scenario-planning platform, and an AI conversational interface for understanding wealth.

Design principle: AI is not the source of financial truth. Portfolio facts come from authorized internal APIs; prices and research come from trusted data providers; calculations come from tested financial services; the LLM explains, compares and orchestrates these results.

2. Product Objectives

- Provide a unified view of investments across countries, currencies and asset classes.
- Support Android and iOS from a shared Flutter codebase with native security capabilities.
- Measure portfolio value, return, income, allocation, concentration and historical growth.
- Provide compound-growth modelling for lump sums, recurring contributions, inflation and scenarios.
- Allow users to define wealth goals and calculate contributions required to reach them.
- Track assets and liabilities to calculate net worth.
- Provide an AI Wealth Copilot for natural-language portfolio questions.
- Provide AI Investment Intelligence that surfaces research opportunities using current evidence, user constraints and portfolio context.
- Maintain security, privacy, auditability and explainability suitable for a financial application.
- Remain extensible for broker/bank integrations, family wealth and adviser capabilities.

3. Mobile Platforms and User Experience

3.1 Android and iOS

Android: Google Play distribution. **iOS:** Apple App Store distribution. Flutter is recommended to share most application code while supporting Face ID, Touch ID, Android biometrics, secure key storage, push notifications and deep links.

3.2 Primary Navigation

Home | Portfolio | AI Wealth | Goals | More

- Home: total wealth, performance, allocation, passive income, goals and AI insight cards.
- Portfolio: portfolios, holdings, transactions, asset detail, performance and allocation.
- AI Wealth: chat, investment opportunities, Portfolio Doctor, research, comparison, simulator and watchlist.
- Goals: target amount, progress, forecast and required contribution.
- More: net worth, calculator, reports, alerts, documents, profile, security and settings.

4. Supported Investment Vehicles

Asset Class	Examples	Measurements
Stocks	Listed equities	Price, value, P/L, dividends, return
ETFs / Funds	Index, thematic, mutual	NAV/price, distributions, return
Bonds	Government/corporate	Coupon, maturity, yield, value
MMF / Fixed Deposit	Cash-management	Principal, yield, interest, maturity
Real Estate / REITs	Property/listed property	Valuation, rent, distributions, appreciation
Crypto	BTC, ETH, others	Quantity, market value, P/L
Pension / SACCO	Retirement/co-operative	Contributions, value, income
Private Business	Private ownership	Valuation, ownership, distributions
Cash / Commodities	Savings, gold, etc.	Balance/quantity, value, return
Custom Assets	User-defined	Configurable valuation and income

5. Core Functional Modules

5.1 Identity & Security

Registration, email verification, MFA, OAuth/OIDC, biometric unlock, profile, base currency, locale, risk profile, investment horizon, notifications and privacy settings.

5.2 Portfolio Management

Multiple portfolios, holdings, valuations, portfolio currency, historical value, notes and documents.

5.3 Transactions

Buy, sell, deposit, withdrawal, dividend, interest, coupon, rent, distribution, fee, tax, transfers and valuation adjustments. Preserve auditability.

5.4 Multi-Currency

Keep investments in native currencies while consolidating reports into the user's base currency using timestamped FX data.

5.5 Portfolio Analytics

Invested capital, market value, realized/unrealized P/L, return, CAGR, XIRR, time-weighted return, income yield, allocation, concentration, currency/geographic/sector exposure, drawdown and volatility.

5.6 Compound Growth & Forecasting

Initial capital, recurring contributions, contribution escalation, expected return, compounding, inflation, fees and time horizon.

5.7 Financial Goals

Target amount/date, progress, gap, required contribution, estimated completion and scenario comparison.

5.8 Passive Income

Dividends, interest, coupons, rent, REIT distributions, business distributions and other recurring investment income.

5.9 Net Worth

Assets less liabilities including property, investments, cash, businesses, mortgages and loans.

5.10 Market Data & Research

Prices, FX, historical data, licensed research/news and macro indicators with provenance and timestamps.

5.11 Watchlists & Alerts

Price targets, maturity reminders, allocation thresholds, contribution reminders and goal milestones.

6. Compound Growth and Forecasting Engine

The calculator must be implemented as deterministic, independently tested financial code rather than an LLM calculation.

Core lump-sum model: $FV = PV \times (1 + r)^n$

- Inputs include starting capital, recurring contribution, expected return, frequency, period, inflation, contribution growth and fees.
- Support conservative, base and aggressive user-defined scenarios.
- Return projected value, total contributed capital, projected growth, inflation-adjusted value and yearly chart series.
- Solve for the periodic contribution required to reach a target.
- Clearly identify assumptions; projected returns are not guaranteed.

7. AI Wealth Copilot

The AI Wealth Copilot provides a conversational interface to authorized portfolio data and deterministic tools. It can answer portfolio questions and invoke forecasting or analytics functions.

Security boundary: the LLM receives no unrestricted database or SQL access. It can only call allow-listed internal tools that validate authorization and return structured results.

8. AI Investment Intelligence

This module identifies possible investment opportunities for research using current market evidence, user constraints and portfolio context. A controlled pipeline filters and scores the investment universe, checks suitability and diversification, retrieves evidence, and then uses the LLM to produce a structured explanation.

8.1 Inputs

- Risk profile, investment horizon, available capital and liquidity needs.
- Existing allocation, concentration, currency and geographic exposure.
- Financial goals.
- Current prices and historical data.
- Fundamental and valuation metrics where applicable.
- Macroeconomic indicators and licensed research/news signals.
- Data freshness and quality.

8.2 Opportunity Output

- Instrument and asset class.
- Thesis and why it surfaced.
- Supporting evidence and data-as-of timestamp.

- Risk level and key downside factors.
- Portfolio and diversification impact.
- Scenario assumptions and research horizon.
- Transparent score explanation if scoring is used.
- Evidence/source references and actions to Research, Compare or Simulate.

Guardrail: opportunity scores are analytical aids, not promises of performance. The LLM must not invent market facts or expected returns.

9. Portfolio Doctor

Portfolio Doctor evaluates concentration, asset-class balance, sector/geographic/currency exposure, liquidity, volatility, income dependence and goal alignment. It produces observations and alternative scenarios rather than automatically executing trades.

10. Technical Architecture

Recommended stack: Flutter/Dart for Android and iOS; Python/FastAPI, Pydantic, SQLAlchemy, Alembic and PostgreSQL for backend services; Redis for caching; RabbitMQ/Celery for asynchronous work; Docker for local/production packaging; Keycloak for open-source identity; OpenAI through a dedicated AI orchestrator.

```
Android / iOS (Flutter) | API Gateway | +-----+-----+-----+ | | Identity Portfolio
Analytics | | +-----+ PostgreSQL -----+ | AI Orchestrator / \ Internal Tools Market/Research \ /
OpenAI / LLM
```

For the MVP, a modular monolith can reduce operational complexity while preserving service boundaries for later extraction.

11. Core Data Model

- users, user_profiles, risk_profiles
- portfolios, portfolio_members
- assets, asset_metadata, holdings
- transactions, valuations, market_prices, fx_rates
- income_events, liabilities
- goals, goal_scenarios, forecast_scenarios
- watchlists, watchlist_items, alerts
- research_items
- ai_conversations, ai_messages, ai_tool_calls
- ai_recommendation_runs, ai_opportunities
- audit_events, documents, notification_preferences

Use UUID identifiers and PostgreSQL NUMERIC/DECIMAL for money and quantities where precision matters. Avoid binary floating-point for money.

12. API and AI Tool Strategy

Use versioned REST endpoints under /api/v1 with OpenAPI contracts generated by FastAPI. The backend remains authoritative for financial calculations.

Allow-listed AI tools should include portfolio summary, allocation, performance, net worth, goal progress, compound forecast, market snapshot, investment-universe search, asset research, asset comparison and portfolio risk analysis. Every tool must validate user authorization and log an auditable event.

13. Security and Privacy Requirements

- TLS 1.2+; prefer TLS 1.3; encryption at rest.
- OAuth/OIDC with PKCE and MFA.
- Secure mobile token storage and refresh-token rotation.
- Server-side authorization on every resource; IDOR tests.
- Rate limiting, brute-force protection and input validation.
- Secrets in Vault/cloud secret manager, never source control.
- Audit trails for financial changes and AI recommendation runs.
- Backups, restore tests and disaster recovery.
- SAST, dependency/container scanning and SBOM generation.
- OWASP MASVS and OWASP API Security reviews.
- Privacy, retention, export and account-deletion workflows.

14. AI Quality and Explainability

- AI is not the financial calculation engine or market-data source.
- Only allow-listed tools access user data.
- Validate LLM output with structured schemas.
- Record model/provider/version and data timestamps.
- Test prompt injection and cross-user data access.
- Qualify stale or insufficient evidence.
- Separate observed data from assumptions/scenarios.
- No automatic trading in the initial release.
- The user makes the final investment decision.

15. Delivery Roadmap

Phase	Primary Deliverables
0 - Foundation	Repository, Flutter shell, FastAPI, PostgreSQL, Redis, Keycloak, Docker, CI/CD, telemetry
1 - Portfolio Core	Portfolios, assets, transactions, holdings, manual valuation, multi-currency dashboard
2 - Analytics	Performance, allocation, income, net worth, calculator, goals
3 - Market Data	Prices, FX, historical data, automatic valuation, watchlists
4 - AI Copilot	AI orchestrator, portfolio tools, secure chat, Portfolio Doctor, audit trail
5 - AI Intelligence	Opportunity screening, evidence, scoring, suitability, compare/simulate
6 - Production	Security/load tests, backup/DR, monitoring, App Store and Google Play releases

16. MVP Acceptance Criteria

- One codebase builds for Android and iOS.
- Secure portfolios and supported investment/transaction recording.
- Backend-authoritative portfolio totals and multi-currency views.
- Deterministic compound and goal forecasts with automated tests.
- Portfolio performance, allocation, income and net worth views.
- Market data with provenance and freshness.
- AI answers authorized portfolio questions only through controlled tools.
- AI opportunities include rationale, risks, assumptions, timestamp and evidence.
- Cross-user data access is prevented and tested.
- CI/CD, logging, monitoring, backups and security scanning operate before production.

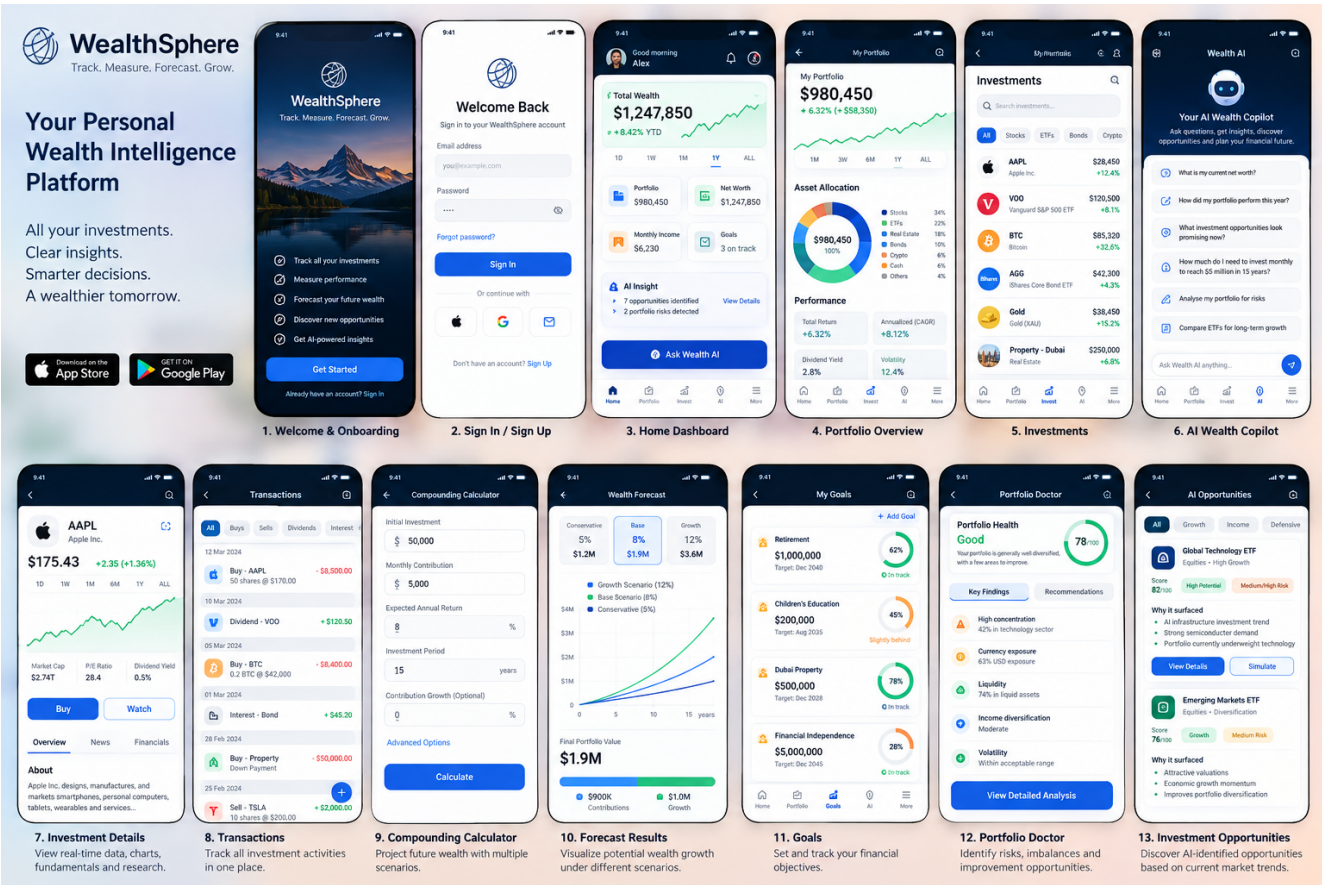
17. Recommended Development Sequence

Repository + CI -> Identity -> Database schema -> Portfolio/assets -> Transactions/holdings -> Multi-currency -> Dashboard -> Analytics -> Compound calculator -> Goals -> Net worth/income -> Market/FX integration -> AI tool layer -> AI Wealth Copilot -> Investment Intelligence -> Production hardening -> Android/iOS store release.

The accompanying Markdown implementation guide is designed for direct use with Claude Code and contains repository structure, data model, API examples, AI tool contracts, security requirements, testing strategy and staged implementation prompts.

Appendix A — Mobile UI/UX Design Specification

WealthSphere shall provide a premium, modern fintech user experience on Android and iOS, with consistent navigation, responsive cards, interactive analytics, AI-assisted research and accessible forecasting. The following visual is a concept reference; all displayed investment values are illustrative.



Design tokens: Navy #102A43 · Teal #0D9488 · Gold #F2B84B · Primary Blue #2563EB. Support light/dark mode, dynamic text, screen readers, semantic status labels and platform-specific navigation conventions.

Primary navigation: Home | Portfolio | AI Wealth | Goals | More

Screen Inventory

Screen	Required experience
Welcome & Onboarding	Brand intro, value proposition, onboarding carousel, Get Started / Sign In, skip accessible where appropriate
Sign In / Sign Up	Email, password or configured identity flow, MFA, secure reset, biometrics for returning users, validation and error feedback
Home Dashboard	Total wealth, return and period selector, chart, portfolio/net worth/income/goal cards, AI insight summary, personalized reorderable widgets
Portfolio Overview	Portfolio selector, value, P/L, line chart, allocation donut and legend, performance metrics, drill-down to holdings
Investments / Holdings	Search, asset-class chips, sortable holdings, native currency and converted value, positive/negative change with accessible text
AI Wealth Copilot	Conversation thread, suggested questions, contextual portfolio scope, source/data timestamps, streaming/loading, refusal and retry states

Screen	Required experience
Investment Details	Symbol/name, latest value, chart and period selector, fundamentals, income, holdings, transaction history, research, watchlist action
Transactions	Filter chips, grouped dated activity, amounts/fees/currencies, add transaction flow, confirmation, edit/correction audit trail
Compounding Calculator	Initial capital, recurring contribution, return assumption, horizon, contribution growth, inflation/fees advanced inputs, Calculate
Wealth Forecast	Conservative/base/growth selectable scenario cards, comparison chart, final value, contributions vs growth, nominal/real toggle, assumptions disclosure
Financial Goals	Goal cards, target and date, progress rings with numeric labels, on-track status, add/edit goal, required contribution
Portfolio Doctor	Health overview, concentration/currency/liquidity/volatility findings, severity + explanation, alternative scenario actions
AI Investment Opportunities	Research category chips, candidate cards, evidence, risk, data-as-of, explainable ranking, Compare/Research/Simulate actions

Reusable components and delivery acceptance

Implement shared wealth cards, portfolio switcher, investment rows, performance/allocation charts, scenario cards, forecast charts, goal progress, health findings, AI opportunity cards, evidence chips, AI chat composer and consistent loading/error/empty states.

Charts must expose accessible summaries and selectable periods. Simulate actions pass clearly marked assumptions to the tested backend forecasting engine. Research actions expose sources and data timestamps. The AI UI must not imply automatic trade execution.

Build design tokens and navigation first; then high-fidelity Home, Portfolio, Calculator, Forecast and AI Wealth screens; then the other eight screens. Verify Android and iOS safe areas, dark mode, keyboard layouts, dynamic type, screen-reader behavior and automated widget tests.